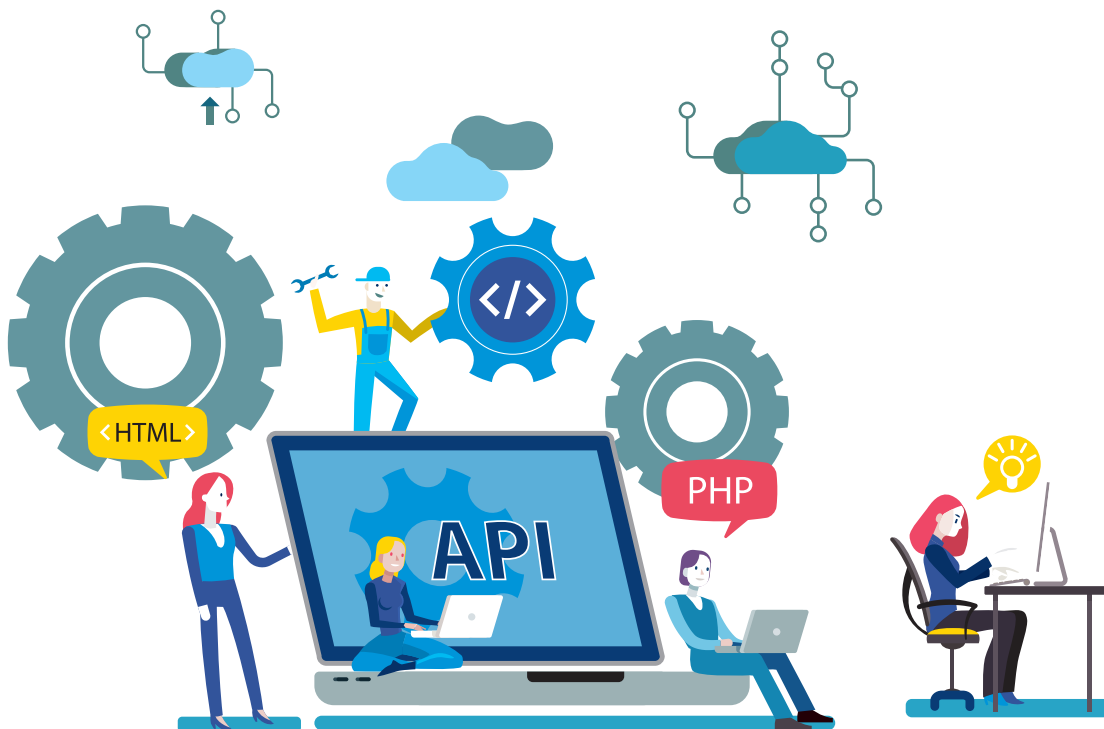


Demystifying Observability for Modern Applications

To come up with an effective solution to a problem, you need to have the relevant insights into it. Distributed systems are prone to unexpected failures and malfunctions, and observability helps us find out the reasons for these in order to arrive at an effective solution.



Operating a distributed system is hard, not just because of its complexity, but also because of the unpredictability of failure modes. With quick software deliveries, continuous build deployments and modern cloud architectures, the possible failure modes are increasing. Unfortunately, traditional monitoring tools cannot help us to deal with these challenges anymore.

What caused these problems to begin with?

Tech giants like Amazon and Microsoft started building their

applications with monolithic architecture as it was more convenient back in the day. They all had to go through the same challenges and hence chose microservices as a better solution, as these can be developed, scaled, and deployed independently. This significantly increased the delivery speed and scalability of applications, but on the flip side, they now have to deal with a different kind of operational complexity related to managing these microservices deployments. One of the benefits of working with older technologies was that only a limited set of failures occurred.

Using application programming interfaces (APIs) to expose core business functionality and facilitating service-to-service communication gives several control points, and makes it easier to build these complex modern applications. However, API-driven applications come with their own set of issues such as design complexity, lack of end-to-end visibility, and security concerns. We are going to discuss that in detail.

A peek into modern problems

Every company or organisation that is adapting to these microservices and